Module B2 — Cells, Chemicals and Transport

2.1 Cell Structure

Eukaryotic Cells

Eukaryotic cells are found in animals, plants, fungi, and protoctists. They are characterised by membrane-bound organelles and a true nucleus containing linear chromosomes enclosed within a nuclear envelope.

Organelle: A discrete structure within a cell, usually membrane-bound, that has a specific function.
Eukaryotic cell: A cell with a true nucleus and membrane-bound organelles.

Organelle	Structure	Function
Nucleus	Double membrane (nuclear envelope) with nuclear pores; contains chromatin and nucleolus	Houses DNA; controls cell activities via mRNA production; nucleolus produces rRNA for ribosome assembly
Rough endoplasmic reticulum (RER)	Flattened membrane sacs (cisternae) studded with ribosomes	Synthesis and transport of proteins, especially secretory proteins
Smooth endoplasmic reticulum (SER)	Tubular membrane network without ribosomes	Lipid synthesis, steroid hormone production, detoxification
Golgi apparatus	Stack of flattened membrane cisternae with vesicles budding off	Modifies, packages, and sorts proteins and lipids; forms lysosomes and secretory vesicles

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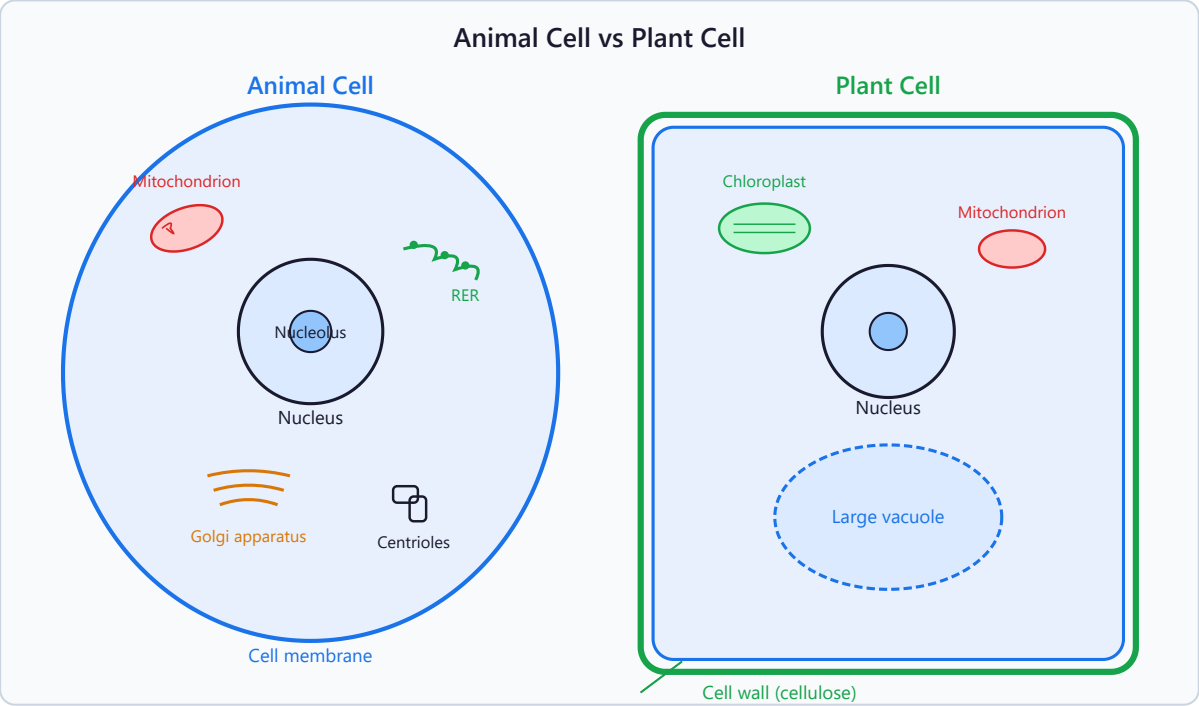
Mitochondria	Double membrane; inner membrane folded into cristae; matrix contains own DNA and ribosomes	Site of aerobic respiration; produces ATP via oxidative phosphorylation
Chloroplasts (plants only)	Double membrane; internal thylakoid membranes arranged in grana; stroma contains own DNA and ribosomes	Site of photosynthesis; light-dependent reactions on thylakoids, Calvin cycle in stroma
Ribosomes	Two subunits (large and small); 80S in eukaryotes	Site of protein synthesis (translation)
Lysosomes	Single membrane vesicle containing hydrolytic enzymes	Intracellular digestion; breakdown of worn-out organelles, pathogens, and cellular debris
Centrioles	Two bundles of nine triplet microtubules arranged at right angles	Organise spindle fibres during cell division



Prokaryotic Cells

Prokaryotic cells (bacteria and archaea) are much smaller than eukaryotic cells (typically 1-5 micrometres) and lack membrane-bound organelles. They have a simpler internal organisation but are highly successful

organisms.

Feature	Prokaryotic Cell	Eukaryotic Cell
Size	1-5 micrometres	10-100 micrometres
Nucleus	No true nucleus; circular DNA in nucleoid region	True nucleus with nuclear envelope
DNA	Single circular chromosome + plasmids; no histones	Linear chromosomes associated with histone proteins
Ribosomes	70S (smaller)	80S (larger)
Organelles	No membrane-bound organelles	Membrane-bound organelles present
Cell wall	Peptidoglycan (murein)	Cellulose (plants), chitin (fungi), or absent (animals)
Reproduction	Binary fission	Mitosis and meiosis

Exam Tip: Be careful with the term "no organelles" for prokaryotes. They do have ribosomes (which are not membrane-bound). The correct statement is that prokaryotes lack membrane-bound organelles.

Microscopy

Light microscopes have a maximum resolution of approximately 200 nm (limited by the wavelength of visible light, ~400-700 nm) and magnification up to about x1500. Electron microscopes use beams of electrons (much shorter wavelength) to achieve resolution down to about 0.5 nm, allowing visualisation of ultrastructure such as ribosomes, membranes, and internal organelle detail.

Feature	Light Microscope	TEM	SEM
Maximum resolution	~200 nm	~0.5 nm	~3-10 nm

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Maximum magnification	~x1500	~x500,000	~x100,000
Image type	2D colour image	2D black/white (thin sections)	3D surface image
Living specimens?	Yes	No (vacuum, staining)	No (vacuum, coating)
Cost/portability	Low cost, portable	Very expensive, large	Expensive, large
Best for	Cell structure, tissues, living processes	Internal ultrastructure of organelles	Surface detail (e.g., insect eyes, pollen)

Preparing slides: Specimens for light microscopy are typically stained to increase contrast (e.g., iodine stains starch blue-black and makes cell walls visible; methylene blue stains nuclei blue; acetic orcein stains chromosomes dark red). For electron microscopy, specimens are fixed (e.g., with glutaraldehyde), dehydrated, embedded in resin, sectioned with an ultramicrotome, and stained with heavy metals (lead citrate, uranyl acetate) that scatter electrons.

$$\text{Magnification} = \text{Image size} / \text{Actual size}$$

2.2 Biological Molecules

Water

Water is the most abundant molecule in living organisms and is essential for life. Its unique properties arise from its polar nature and ability to form hydrogen bonds.

- **Solvent:** Polarity allows ionic and polar substances to dissolve, making water an excellent transport medium and reaction medium
- **High specific heat capacity:** Large amounts of energy needed to raise temperature; buffers organisms against rapid temperature changes
- **High latent heat of vaporisation:** Evaporation provides effective cooling (sweating, transpiration)
- **Cohesion and adhesion:** Hydrogen bonds create surface tension and allow water to travel up xylem vessels against gravity

- **Maximum density at 4 degrees C:** Ice floats, insulating water below and allowing aquatic life to survive in winter

Carbohydrates

Carbohydrates contain carbon, hydrogen, and oxygen in the ratio Cn(H2O)n. They serve as energy sources, energy stores, and structural components.

Monosaccharides are the simplest sugars (e.g., glucose, fructose, galactose). Glucose exists as alpha and beta isomers, which differ in the position of the hydroxyl group on carbon-1. This seemingly small difference leads to vastly different polysaccharide structures.

Disaccharides are formed when two monosaccharides join via a condensation reaction, forming a glycosidic bond and releasing water. Maltose (glucose + glucose), sucrose (glucose + fructose), and lactose (glucose + galactose) are common examples.

Polysaccharides are polymers of many monosaccharides:

- **Starch:** Energy store in plants. Contains amylose (unbranched, alpha-1,4 glycosidic bonds, helical) and amylopectin (branched, alpha-1,4 and alpha-1,6 bonds). Compact and insoluble, so does not affect water potential.
- **Glycogen:** Energy store in animals. Similar to amylopectin but more highly branched, allowing rapid hydrolysis for quick energy release.
- **Cellulose:** Structural component of plant cell walls. Beta-glucose monomers linked by beta-1,4 glycosidic bonds; adjacent chains form hydrogen bonds creating strong microfibrils.

Lipids

Lipids are a diverse group of hydrophobic molecules. Triglycerides consist of one glycerol molecule esterified to three fatty acid chains via condensation reactions forming ester bonds.

Fatty acids can be **saturated** (no C=C double bonds, straight chains, solid at room temperature) or **unsaturated** (one or more C=C double bonds, kinked chains, liquid at room temperature). The kinks prevent close packing.

Phospholipids have two fatty acid tails and a phosphate head group. The head is hydrophilic (polar) while the tails are hydrophobic (non-polar), making phospholipids amphipathic. This property drives the spontaneous formation of bilayers in aqueous environments.

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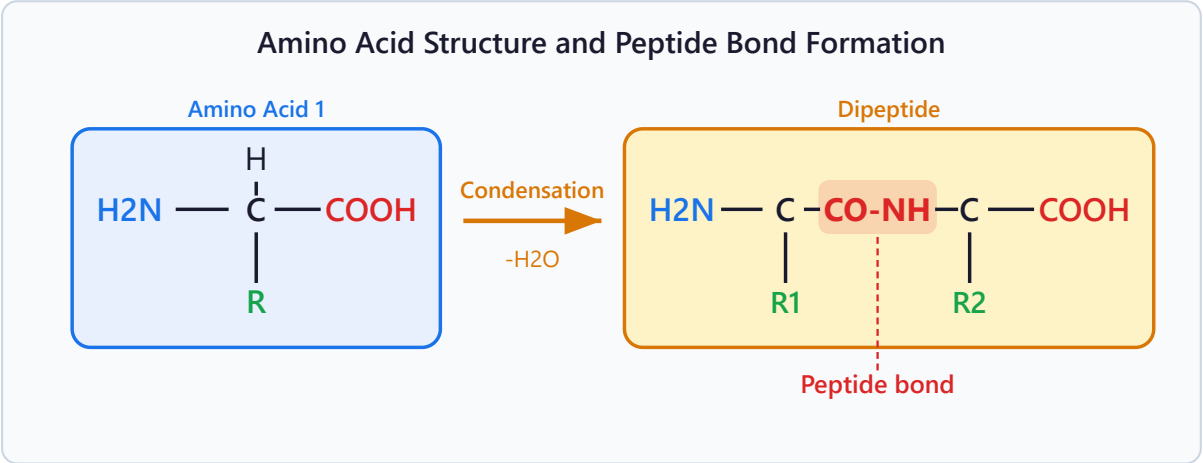
Condensation reaction: A chemical reaction in which two molecules join together with the removal of water.
Hydrolysis: The breaking of a chemical bond by the addition of water.

Proteins

Proteins are polymers of amino acids joined by peptide bonds formed through condensation reactions. There are 20 different amino acids, each with a unique R-group that determines its properties.



Protein structure is organised into four levels:

- **Primary structure:** The sequence of amino acids in the polypeptide chain, determined by DNA
- **Secondary structure:** Local folding into alpha-helices and beta-pleated sheets, stabilised by hydrogen bonds between C=O and N-H groups of the peptide backbone
- **Tertiary structure:** The overall 3D shape of a single polypeptide, maintained by hydrogen bonds, ionic bonds, disulfide bridges, and hydrophobic interactions between R-groups
- **Quaternary structure:** The arrangement of multiple polypeptide subunits and any prosthetic groups (e.g., haemoglobin has four subunits plus haem groups)

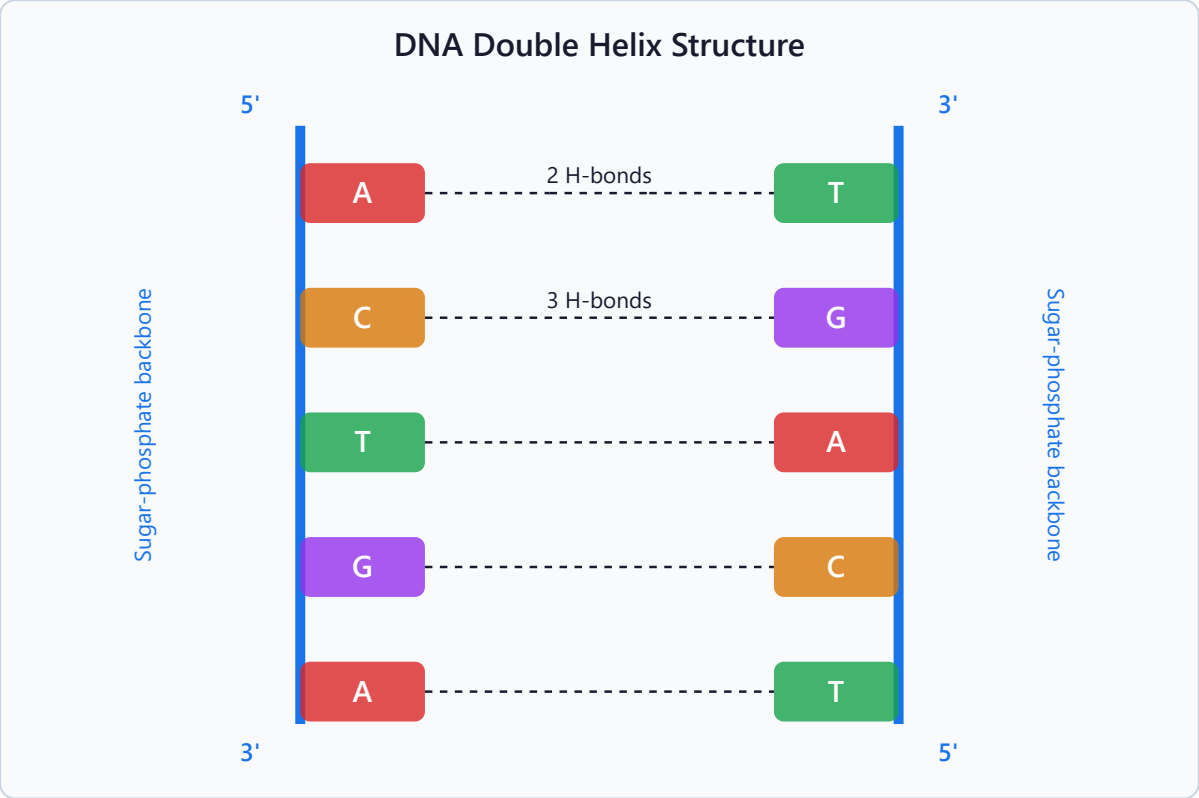
Proteins can be **globular** (soluble, spherical, metabolic roles such as enzymes and antibodies) or **fibrous** (insoluble, elongated, structural roles such as collagen and keratin).

Nucleic Acids

DNA and RNA are polynucleotides. Each nucleotide consists of a pentose sugar, a phosphate group, and a nitrogenous base. In DNA the sugar is deoxyribose and bases are A, T, C, G; in RNA the sugar is ribose and

bases are A, U, C, G.

DNA forms a double helix with antiparallel strands held together by complementary base pairing (A-T with two hydrogen bonds, C-G with three hydrogen bonds). The sugar-phosphate backbone provides structural stability while the base sequence encodes genetic information.



2.3 Enzymes

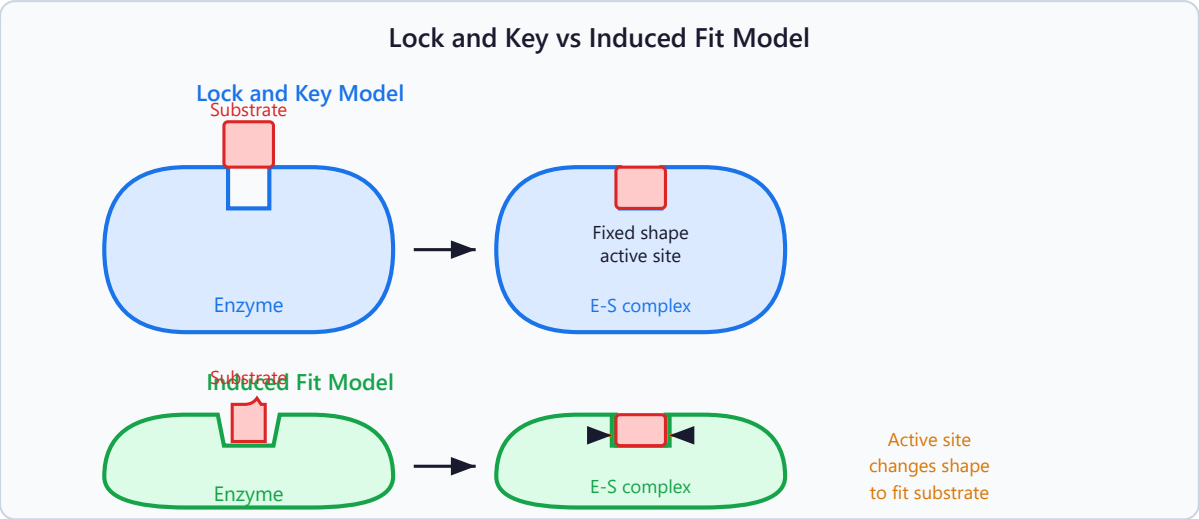
Enzymes are biological catalysts, usually globular proteins, that speed up metabolic reactions by lowering the activation energy. They are highly specific due to the precise shape of their active site.

Active site: A region on the surface of an enzyme with a specific shape complementary to the substrate. This is where the substrate binds and the reaction is catalysed.
Activation energy: The minimum energy required for a reaction to occur.

Models of Enzyme Action

The **lock and key model** proposes that the active site has a fixed shape exactly complementary to the substrate. The **induced fit model** (more accepted) proposes that the active site changes shape slightly as

the substrate binds, moulding around it to form a tighter fit. This conformational change puts strain on bonds in the substrate, lowering activation energy further.



Factors Affecting Enzyme Activity

- **Temperature:** Increasing temperature increases kinetic energy and collision frequency, raising the rate up to the optimum. Beyond the optimum, hydrogen bonds and other weak interactions maintaining tertiary structure break, the active site changes shape (denaturation), and rate drops to zero.
- **pH:** Changes in pH alter the ionisation of R-groups, disrupting ionic bonds and hydrogen bonds in the active site. Each enzyme has an optimum pH.
- **Substrate concentration:** At low concentrations, increasing substrate increases rate (more successful collisions). At high concentrations, all active sites are occupied (enzyme saturation) and rate plateaus. Increasing enzyme concentration raises the maximum rate.

Inhibitors

Competitive inhibitors have a similar shape to the substrate and bind to the active site, blocking substrate access. Their effect can be overcome by increasing substrate concentration. **Non-competitive inhibitors** bind to an allosteric site (elsewhere on the enzyme), causing a conformational change that alters the active site shape so substrate can no longer bind effectively. Increasing substrate concentration does not overcome this.

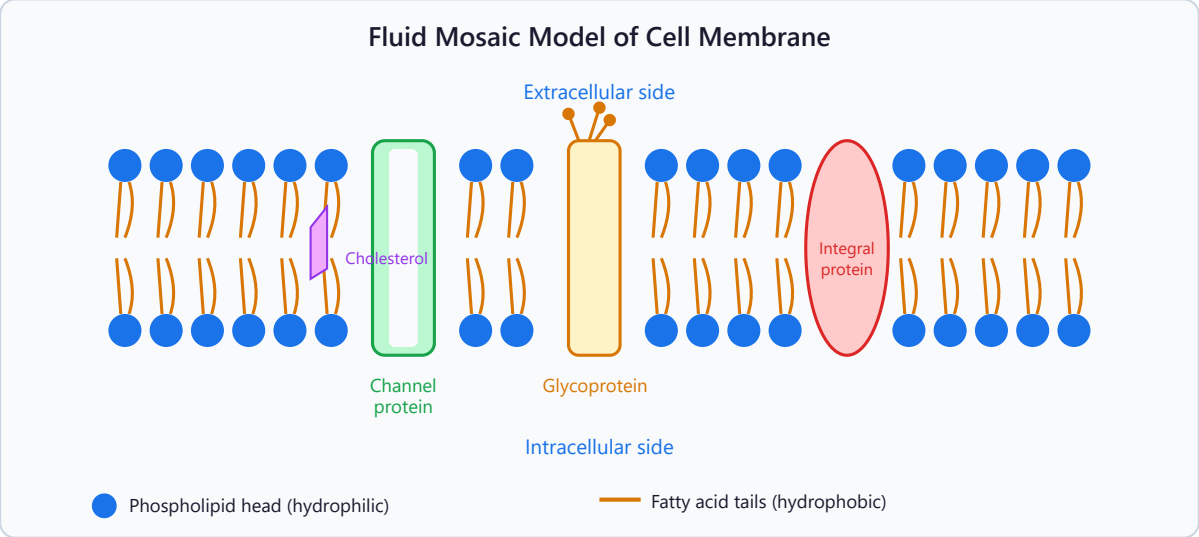
2.4 Cell Membranes

The Fluid Mosaic Model

Cell membranes are composed of a phospholipid bilayer with proteins embedded in or attached to it. The model is called "fluid" because phospholipids and proteins can move laterally within the bilayer, and "mosaic" because of the varied pattern of proteins scattered throughout.

Components of the membrane include:

- **Phospholipid bilayer:** Forms the basic structure; hydrophobic core acts as barrier to polar and charged molecules
- **Cholesterol:** Sits between phospholipid tails, regulating fluidity (reduces at high temperatures, prevents solidifying at low temperatures)
- **Integral (transmembrane) proteins:** Span the entire bilayer; function as channels, carriers, or receptors
- **Peripheral proteins:** Attached to inner or outer surface; often enzymes or involved in cell signalling
- **Glycoproteins and glycolipids:** Carbohydrate chains attached to proteins or lipids on the outer surface; involved in cell recognition, signalling, and immune response



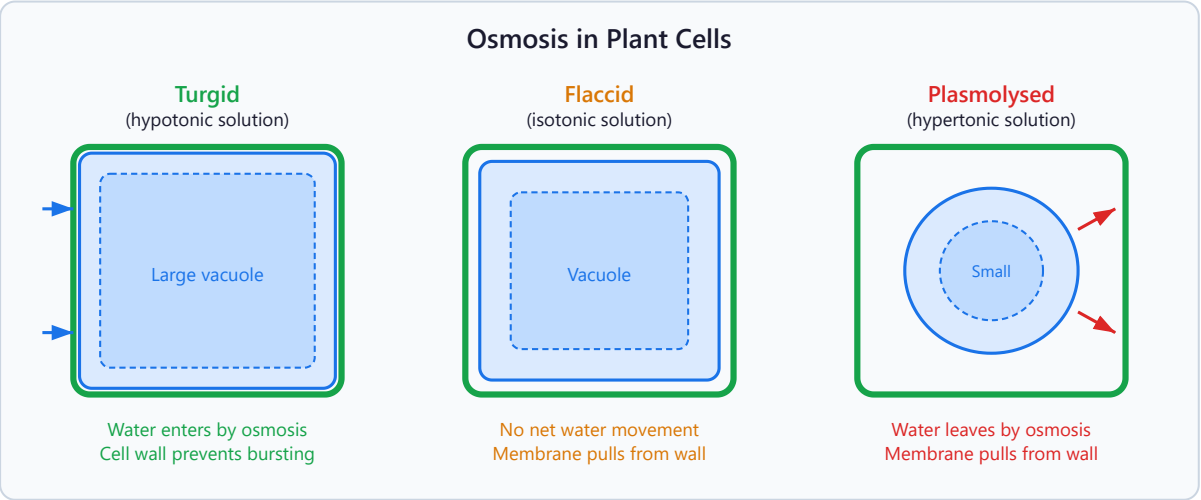
Transport Across Membranes

Mechanism	Description	Energy Required?
Diffusion	Net movement of molecules from high to low concentration down a concentration gradient	No (passive)

Facilitated diffusion	Diffusion of polar/charged molecules through channel proteins or carrier proteins	No (passive)
Osmosis	Net movement of water molecules from high water potential to low water potential across a partially permeable membrane	No (passive)
Active transport	Movement of molecules against concentration gradient using carrier proteins and ATP	Yes (ATP)
Endocytosis	Bulk transport of large molecules into the cell by membrane invagination forming vesicles	Yes (ATP)
Exocytosis	Bulk transport of large molecules out of the cell by vesicles fusing with the membrane	Yes (ATP)

Water potential (psi): A measure of the tendency of water molecules to move from one region to another. Pure water has a water potential of zero; dissolved solutes lower water potential to negative values. Water always moves from higher (less negative) to lower (more negative) water potential.

Exam Tip: Never say water moves from "high concentration to low concentration." The correct terminology is that water moves down a water potential gradient, from a region of higher (less negative) water potential to a region of lower (more negative) water potential.



Factors Affecting Membrane Permeability

Temperature affects membrane permeability: at low temperatures, phospholipids pack closely and the membrane becomes rigid and less permeable. As temperature increases, fluidity increases and permeability rises. Above approximately 40 degrees C, proteins denature and the phospholipid bilayer becomes highly disrupted, causing a large increase in permeability. Solvents such as ethanol dissolve membrane lipids, also increasing permeability.

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